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Coptidis rhizoma extract alleviates oropharyngeal candidiasis by gC1qR-EGFR/ERK/c-fos axis-induced endocytosis of oral epithelial cells

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A R T I C L E I N F O

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A B S T R A C T

Ethnopharmacological relevance: Coptidis rhizoma, first recorded in the "Shen Nong's Herbal Classic", is one of the traditional Chinese medicine (TCM) used to treat infectious diseases, with reputed effectiveness against oropharyngeal candidiasis (OPC). Studies have demonstrated the inhibitory properties of *C. rhizoma* (CRE) against *Candida albicans*, yet there is limited information available regarding its treatment mechanism for OPC.
Aim of the study: Our previous research has suggested that CRE can prevent the formation of *C. albicans* hyphae and their invasion of the oral mucosa, thereby exerting a therapeutic effect on OPC. Nevertheless, the precise therapeutic mechanisms remain incompletely understood. Previous studies have revealed that a receptor for globular heads of C1q (gC1qR), a crucial co-receptor of the epidermal growth factor receptor (EGFR), facilitates the EGFR-mediated internalization of *C. albicans*. Therefore, this study aims to investigate the potential mechanism of action of CRE and its primary component, berberine (BBR), in treating OPC by exploring their effects on the gC1qR-EGFR co-receptor.
Materials and methods: To identify the chemical components of CRE, we utilized Ultra-high performance liquid chromatography in conjunction with quadrupole time-of-flight mass spectrometry (UPLC-Q/TOF-MS^E), revealing the presence of at least 18 distinct components. To observe the therapeutic effects of CRE on OPC at the animal level, we employed hematoxylin and eosin staining, periodic acid-Schiff staining, scanning electron microscopy, and fungal load detection. Subsequently, we evaluated the anti-inflammatory properties of CRE and its main component, BBR, in treating OPC. This was achieved through enzyme-linked immunosorbent assay (ELISA) both at the animal and cellular levels. Additionally, we assessed the ability of *C. albicans* to disrupt the epithelial barrier of FaDu cells by studying the protective effects of BBR on the fusion barrier using the transwell assay. To further explore the underlying mechanisms, we analyzed the effects of BBR on the gC1qR-EGFR/extracellular signal-regulated kinase/c-Fos signaling pathway at the cellular level using qRT-PCR, western blotting, and immunofluorescence. Furthermore, we validated the effects of BBR on the gC1qR-EGFR co-receptor through ELISA, qRT-PCR, and western blotting. Finally, to confirm the outcomes observed at the cellular level, we validated the impact of CRE on the gC1qR-EGFR co-receptor in vivo using qRT-PCR, western blotting, and immunofluorescence. These comprehensive methods allowed us to gain a deeper understanding of the therapeutic mechanisms of CRE and BBR in treating OPC.
Results: Our findings indicate that CRE and its primary component, BBR, effectively alleviated the symptoms of OPC by modulating the gC1qR-EGFR co-receptor. The chemical composition of CRE and BBR was accurately identified using UPLC-Q/TOF-MS^E. The gC1qR-EGFR co-receptor plays a crucial role in regulating downstream signaling pathways, emerging as a potential therapeutic target for OPC treatment. Through both *in vitro* and in vivo experiments, we explored the therapeutic potential of CRE and BBR in OPC. Additionally, we employed

Abbreviations: CA, *C. albicans, Candida albicans*; CRE, Coptidis rhizome; BBR, berberine; NYS, nystatin; OPC, oropharyngeal candidiasis; PAS, periodic acid-Schiff; H&E, hematoxylin-eosin staining; PBS, phosphate-buffered saline; YPD, yeast extract peptone dextrose medium; qRT–PCR, quantitative real-time PCR; CFU, colony forming units; ELISA, Enzyme-linked immunosorbent assay.

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overexpression and silencing techniques to confirm that BBR can indeed influence the gC1qR-EGFR co-receptor and regulate the gC1qR-EGFR/extracellular signal-regulated kinase (ERK)/c-Fos signaling pathway, leading to improved OPC outcomes. Furthermore, the significance of CRE's effect on the gC1qR-EGFR co-receptor was validated in vivo.

Conclusion: Our study demonstrates that CRE and its main component, BBR, can effectively alleviate OPC symptoms by targeting the gC1qR-EGFR heterodimer receptor. This discovery offers a promising new therapeutic approach for the treatment of OPC.

1. Introduction

Oropharyngeal candidiasis (OPC), commonly known as mycotic stomatitis, is an infectious fungal disease affecting the dorsal tongue mucous layer. This condition is marked by the extensive damage of lingual papilla, the emergence of white pseudomembranes on the tongue mucosa, or angular cheilitis (Hu et al., 2019; Wang et al., 2023). *Candida albicans* (*C. albicans*) infections, which detach from the oral mucosa, have been observed in 50%–65% of individuals wearing dentures, 65%–88% of those residing in long-term care facilities, 90% of patients with acute leucocythemia undergoing chemotherapy, and 95% of HIV-positive patients (Qadir et al., 2023; Seoudi et al., 2015). OPC can lead to taste disturbances and a burning sensation in the mouth, resulting in various oral discomforts (Rajadurai et al., 2021). The excessive proliferation of *C. albicans* causes mucosal infections, particularly in the superficial layer of the oral mucosa, significantly compromising quality of life and overall health (Ponde et al., 2022; Verma et al., 2017).

C. albicans is a dimorphic yeast fungus, renowned for its virulence capabilities, which encompass adhesion, phenotypic switching, invasion, and biofilm formation. Its transition from a harmless commensal organism to a pathogenic threat is primarily attributed to its ability to alternate between yeast and hyphal forms. This morphological shift plays a pivotal role in mucosal infections, as the hyphae invade and breach the mucosal barrier (Höfs et al., 2016; Lu et al., 2014). During infection, *C. albicans* hyphae secrete a cytotoxic peptide called candidalysin (Clys) onto the mucosal surface. This peptide is crucial for pathogenesis as it triggers calcium ion influx into epithelial cells, activates matrix metalloproteinases, and leads to the shedding of natural epidermal growth factor receptor (EGFR) ligands (Moyes et al., 2016). This shedding process sensitizes the EGFR signaling pathway and triggers a pro-inflammatory response in oral epithelial cells (Ponde et al., 2022). Furthermore, research has revealed that a receptor for globular heads of C1q (gC1qR), a vital co-receptor of EGFR, plays a prominent role in *C. albicans*-induced activation of EGFR receptors. This activation prompts the endocytosis of *C. albicans* and stimulates the secretion of pro-inflammatory mediators (Phan et al., 2021). Interestingly, the agglutinin-like sequence protein 3 (Als3) of *C. albicans* hyphae requires gC1qR to interact with EGFR efficiently and activate it. Additionally, EGFR is essential for optimal interactions between *C. albicans* and gC1qR (Phan et al., 2021). Therefore, inhibiting the activation of the gC1qR-EGFR co-receptor and reducing the production of pro-inflammatory mediators emerge as key therapeutic strategies for treating OPC.

The primary therapeutic approach for OPC involves antifungal medication, including both topical antifungal therapy (such as nystatin, miconazole) and systemic antifungal therapy (such as fluconazole, itraconazole). These medications function by suppressing *C. albicans* to alleviate the associated disorders (Rajadurai et al., 2021; Vila et al., 2020). However, OPC often manifests with increased inflammation and immune system breakdown, worsening the condition. The limited effectiveness of current antifungal therapies in meeting the necessary treatment standards underscores the urgent need for innovative medications. Plants possess numerous pharmacological properties, including immunomodulatory, anti-inflammatory, and antifungal effects, which can be harnessed to reduce inflammation, modulate the host immune system, and suppress *C. albicans*, thereby controlling fungal infections. Studies have shown promising therapeutic potential in plant extracts against disorders associated with *C. albicans* (Authier et al., 2022; Hu et al., 2023; Rusda et al., 2020). The n-butanol extract of *Pulsatilla* decoction exhibits inhibitory effects on *C. albicans*, mitigates damage to macrophages infected with this fungus, prevents the activation of the NLRP3 inflammasome, and modifies host immunity (Hu et al., 2023). Additionally, hydroethanolic extracts derived from licorice root and walnut leaf have been found to reduce colonic inflammation and oxidative stress, regulate intestinal microbiota, and prevent *C. albicans* from colonizing the gastrointestinal tract (Authier et al., 2022). Furthermore, Nigella sativa extract possesses antifungal properties that reduce the number of *C. albicans* colonies and elevate IgM levels in the blood (Rusda et al., 2020). However, the investigation by which plant extracts inhibits *C. albicans* for treating OPC have received little attention.

Coptidis rhizoma, a traditional Chinese medicinal herb with a century-old history, is widely used in daily life to dispel heat and purge toxins from the body. According to the ancient medical text "*Mingyi Bie Lu*," this herb is effective in treating oral ulcers. *Coptidis rhizoma* is derived from the dried rhizomes of *Coptidis rhizoma* Franch. And its botanical name has been verified through the online resource http://www.worldfloraonline.org (accessed on March 11, 2024). This herb is widely distributed across China and is frequently combined with other medications to treat oral ulcers and oral mucositis (Geng et al., 2021). Scientific research has demonstrated that berberine (BBR), a key compound found in high concentrations in *Coptidis rhizoma*, can effectively treat oral candidiasis by inhibiting the formation of *C. albicans* hyphae and preventing their invasion of the oral mucosa (Niu et al., 2019). Furthermore, topical application of BBR gel has been shown to promote ulcer healing and provide relief for recurrent aphthous stomatitis (Jiang et al., 2013). The present study has uncovered that CRE, along with its key component BBR, may effectively suppress *C. albicans* infection and concurrently reduce inflammation-associated infection levels in OPC mice. However, further investigation is necessary to elucidate the precise mechanism underlying these effects.

The co-receptor gC1qR-EGFR plays a critical role in activating EGFR, triggering endocytosis, and promoting the secretion of pro-inflammatory mediators in *C. albicans*. It is also regulated by the EGFR/extracellular signal-regulated kinase (ERK)/c-Fos signaling pathway (Ponde et al., 2022). Therefore, our aim was to explore the mechanism by which CRE and its primary component, BBR, mediate the endocytosis of oral epithelial cells in response to *C. albicans* infection, thereby enhancing the gC1qR-EGFR/ERK/c-Fos axis.

2. Materials and methods

2.1. Ethics statement

All experimental procedures involving animals were approved by the Animal Ethics Committee of the Anhui University of Traditional Chinese Medicine (Animal Ethics Number AHUCM-mouse-2023042). All the animals were maintained and treated in accordance with the principles of the Animal Ethics Committee of the Chinese Center for Disease Control and Prevention and the Guidelines for the Care and Use of Laboratory Animals in China.

2.2. Chemicals and reagents

The *C. rhizoma*, originating from Leshan, Sichuan Province, was procured from the First Affiliated Hospital of Anhui University of Traditional Chinese Medicine (Lot. 24010609). BBR (purity >98%, Lot. 2086-83-1) was purchased from Chengdu Desite Biological Technology Co. Ltd. (Chengdu, China). Nystatin (NYS; 98% > purity) was purchased from Solarbio (Beijing, China). C-Fos, gC1qR, and ERK1/2 were purchased from HUA BIO (ET1701-95, HA601067, and ET1601-29, Hangzhou, China). EGFR, β-Actin Mouse mAb, Goat Anti-Mouse IgG H&L (AF488), and Goat Anti-Rabbit IgG H&L (AF594) were purchased from ZEN BIO (R22778, 200068-8F10, 550036, 550043, Chengdu, China). ERK1/2 (phospho Thr202/Y204) was purchased from ImmunoWay Biotechnology Co., Ltd. (YP1197, Texas, USA). Phospho-EGF Receptor (Tyr1068) was purchased from Cell Signaling Technology (3777, Massachusetts, USA). EGFR, gC1qR, horseradish peroxidase-labeled goat anti-rabbit lgG, CY3-labeled goat anti-rabbit IgG were purchased from ServiceBio (GB111504, GB113769, GB23204, and GB21404, Wuhan, China).

2.3. CRE preparation

The dried *C. rhizoma* plant material (500 g) was finely ground into a powder. This powder was then soaked in 80% ethanol at an 8:1 ratio (ethanol: powder = 8:1) overnight for extraction using a reflux condenser and rotary evaporator. The resulting CRE dry powder was obtained through the process of lyophilization.

2.4. Strains and cultivation

C. albicans SC5314 was generously provided by Professor Jiang Yuan-ying from the School of Pharmacy at the Second Military University in China. The strain was initially inoculated onto YPD agar plates sourced from Hope Biotech Co. in Qingdao, China, and evenly spread across the plates after 24 h. A single colony was selected and thawed in liquid YPD medium from Hope Biotech Co. for 14 h to ensure that the *C. albicans* entered the exponential growth phase, followed by placement in a 37 °C incubator. Subsequently, the *C. albicans* culture was resuspended in serum-free Dulbecco's Modified Eagle Medium from Spark Jade in Shandong, China, and centrifuged at 3000×g for 5 min. Cell counting was performed using a hemocytometer prior to the commencement of the experiments.

2.5. Ultra-high performance liquid chromatography (UPLC)-quadrupole time-of-flight (Q/TOF)- mass spectrometry (MS^E) conditions

The constituents of CRE were analyzed using UPLC-Q/TOF-MS^E, following previously established protocols (Hao et al., 2020; Qin et al., 2018). For the UPLC chromatographic analysis, the ACQUITY UPLC BEH C18 column (100 mm × 2.1 mm, 1.7 μm) sourced from Waters Corporation (Milford, MA, USA) was employed, with a flow rate of 0.2 mL/min and a sample injection volume of 2 μL. The mobile phase comprised of two solutions: solution A (0.1% formic acid in water, v/v) and solution B (0.1% formic acid in acetonitrile, v/v). A gradient elution was performed according to the following schedule: an initial 8% B increasing to 46% B over 0–19 min, followed by a rise to 60% B from 19 to 22 min, a subsequent decrease to 8% B between 22 and 23 min, and a final maintenance at 8% B for 23–25 min.

For mass spectrometry, an electric spray ion source (ESI) was utilized in positive ion mode, with a data acquisition range of 50–1200 m/z. The cone gas flow was set at 50 L/h, while the desolvation gas flow rate was maintained at 600 L/h. The ion source temperature was set to 120 °C, and the desolvation temperature was adjusted to 350 °C. Cone voltage was set at 40 V, and capillary voltage was set at 3.0 kV for ESI^+. MSE mode was chosen, employing a low collision energy of 6 eV and a high collision energy ranging from 20 to 30 eV. Leucine enkephalin (m/z 554.2620) served as an external standard (Lock Spray™) for real-time quality correction, injected at a constant flow rate of 5 μL/min.

2.6. OPC model

The OPC model utilized in this study is primarily based on existing literature on OPC, with some necessary modifications (Chen et al., 2022; Conti et al., 2014; Solis and Filler, 2012; Wang et al., 2023). Sixty female C57BL/6 mice aged 7 weeks old, weighing 20–22 g, were randomly assigned to each group, with 10 mice per group. The mice were acclimatized for one week, provided with ultrapure water, and provided with autoclaved feed and bedding. On days −1, +1, and +3 before and after infection with *C. albicans* SC5314, 225 mg/kg cortisone acetate was injected intraperitoneally for immunosuppression. On day 0, the anesthetized mice were administered 100 mg/mL of urethane. Calcium alginate swabs impregnated with *C. albicans* SC5314 (1×10^6 cells/mL) were placed on the murine tongue mucosa and held for 75 min. Following the infection, the mice were orally administered 50, 100, and 200 mg/kg of CRE on days +1 to +4, with NYS (10 mg/kg) used as a positive control. The mice in the sham group were injected with equal volumes of sterile distilled water. On day +5 post-infection, the mice were euthanized and tissues including the tongue, liver, stomach, lungs, spleen, kidneys, and colon were collected for subsequent experiments.

2.7. Cell culture

The FaDu oropharyngeal squamous cell line was obtained from iCell Bioscience Inc. in Shanghai, China. The FaDu cells were maintained in Dulbecco's Modified Eagle Medium and incubated in a 37 °C, 5% CO_2 environment.

2.8. Fungal burden

Five days after being infected with *C. albicans*, the entire oral cavity of the mice was swabbed daily at the same time using a sterile cotton swab to collect *C. albicans*. The cotton swab with *C. albicans* was then placed in an EP tube, and 0.99 ml of sterile phosphate-buffered saline (PBS) was added. After vortexing for 2–3 min to ensure even mixing of *C. albicans*, the suspension was diluted 10-fold with sterile PBS. From this diluted suspension, 100 μL was smeared onto YPD agar medium supplemented with 100 U/mL penicillin and 0.1 mg/mL streptomycin (double antibiotics), and incubated at 37 °C for 48 h. Following euthanasia, the organs were removed, weighed, and homogenized. The homogenate was diluted 10-fold and then smeared onto YPD agar medium containing double antibiotics. The plates were incubated at 37 °C for 48 h. Fungal colony counts were determined as colony-forming units (CFU) per milliliter or gram.

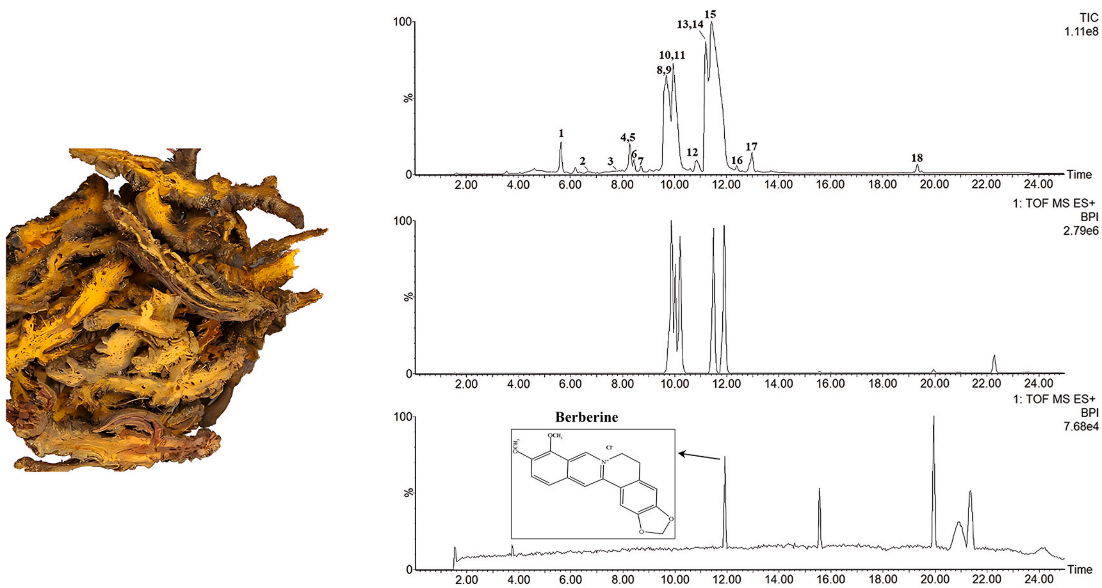
2.9. Hematoxylin and eosin (H&E) and periodic acid-schiff (PAS) staining

The tongue tissue was fixed in 10% paraformaldehyde, embedded in paraffin, and cut into 5-μm thick sections. H&E and PAS staining kits (Servicebio, Wuhan, China) were used according to the manufacturer's instructions to carry out the experiments. The tongue tissue infection was observed under a Nikon upright light microscope at × 200 magnification.

2.10. Scanning electron microscope (SEM)

The fresh tongue tissue was sampled, and any blood stains and surface hair were rinsed with PBS and gently dried on filter paper. The samples were then quickly immersed in an electron microscopy fixative solution at room temperature for 2 h. After fixation and dehydration, the samples were dried in a critical-point desiccator. Following this, the samples were observed using a HITACHI scanning electron microscope

A B



C

No	Compounds	t_R (min)	Formula	Observed mass (Da)	Calculated mass (Da)	Mass error (ppm)	Adducts	MS/MS
1	Magnolflorine	5.65	$C_{20}H_{24}NO_4$	342.1690	342.1700	-2.9	$[M+H]^+$	297.1115、282.0882、265.0875、219.0809
2	Lycoranine	6.67	$C_{15}H_{25}N_2O$	308.0912	308.0923	-3.5	$[M+H]^+$	280.0985、265.0742、250.0885
3	13 - methyl - corydalis Phenol Base	7.79	$C_{21}H_{25}NO_4$	356.1847	356.1862	-4.2	$[M+H]^+$	309.1021、206.1198
4	Stecepharine	8.18	$C_{21}H_{26}NO_5$	372.1811	372.1811	0.0	$[M+H]^+$	342.1842、338.1388
5	Berberubine	8.29	$C_{19}H_{16}NO_4$	322.1062	322.1074	-3.7	$[M+H]^+$	307.0865、279.0893、250.0885
6	Demethyleneberberine	8.44	$C_{19}H_{18}NO_4$	324.1214	324.1236	-6.7	$[M+H]^+$	309.0976、308.0912、294.0791、280.0985、266.0811
7	Demethyleneberberine	8.71	$C_{20}H_{18}NO_5$	352.1186	352.1179	1.9	$[M+H]^+$	337.0937、320.0933、308.0912
8	Demethyleneberberine[1)]	9.60	$C_{20}H_{20}NO_4$	338.1388	338.1392	-1.1	$[M+H]^+$	323.1131、322.1062、320.0933、308.0912、306.1120、294.1142、279.0893、265.0742
9	Coptisine[1)]	9.70	$C_{19}H_{14}NO_4$	320.0897	320.0923	-8.1	$[M+H]^+$	318.0758、292.0965、277.0728、262.0845、249.0767、234.0902、204.0803
10	Columbamine	9.92	$C_{20}H_{20}NO_4$	338.1388	338.1392	-1.1	$[M+H]^+$	323.1131、320.0892、308.0912、294.1107、279.0893、265.0742
11	Epiberberine[1)]	9.99	$C_{20}H_{18}NO_4$	336.1214	336.1236	-6.5	$[M+H]^+$	320.0897、292.0965、264.1024
12	Dehydrocheilanthifoline	10.94	$C_{19}H_{16}NO_4$	322.1062	322.1074	-3.7	$[M+H]^+$	307.0829、279.0893
13	Palmatine[1)]	11.11	$C_{21}H_{22}NO_4$	352.1532	352.1543	-3.1	$[M+H]^+$	336.1214、322.1062、308.1271、294.1142
14	Worenine	11.18	$C_{20}H_{16}NO_4$	334.1063	334.1079	-4.7	$[M+H]^+$	304.0963、278.0818
15	Berberine[1)]	11.40	$C_{20}H_{18}NO_4$	336.1214	336.1236	-6.5	$[M+H]^+$	322.1025、321.0971、306.0762、304.0963、292.0965、278.0818
16	13 - methyl - l - tetrahydropalmatine	12.39	$C_{22}H_{24}NO_4$	366.1701	366.1705	-1.0	$[M+H]^+$	351.1441、350.1364、322.1429、292.0965
17	13 - methyl - table of Berberine	12.98	$C_{21}H_{20}NO_4$	350.1403	350.1392	3.1	$[M+H]^+$	335.1131、334.1100、306.1155、292.1000
18	Oxyberberine	19.33	$C_{20}H_{18}NO_5$	352.1186	352.1179	1.9	$[M+H]^+$	337.0974、322.0731、308.0948、294.0791

Fig. 1. Components present in CRE. (A) *Coptidis rhizoma* (B) The upper panel represents the chemical composition of CRE identified using UPLC, the middle panel represents the mixed control, and the down panel represents the berberine blank control (C) UPLC-Q/TOF-MS^E mass spectrometry data and identification results of the chemical composition of CRE. Compounds (demethyleneberberine, coptisine, epiberberine, palmatine and berberine) labeled with "1)" were confirmed by comparison of reference materials.

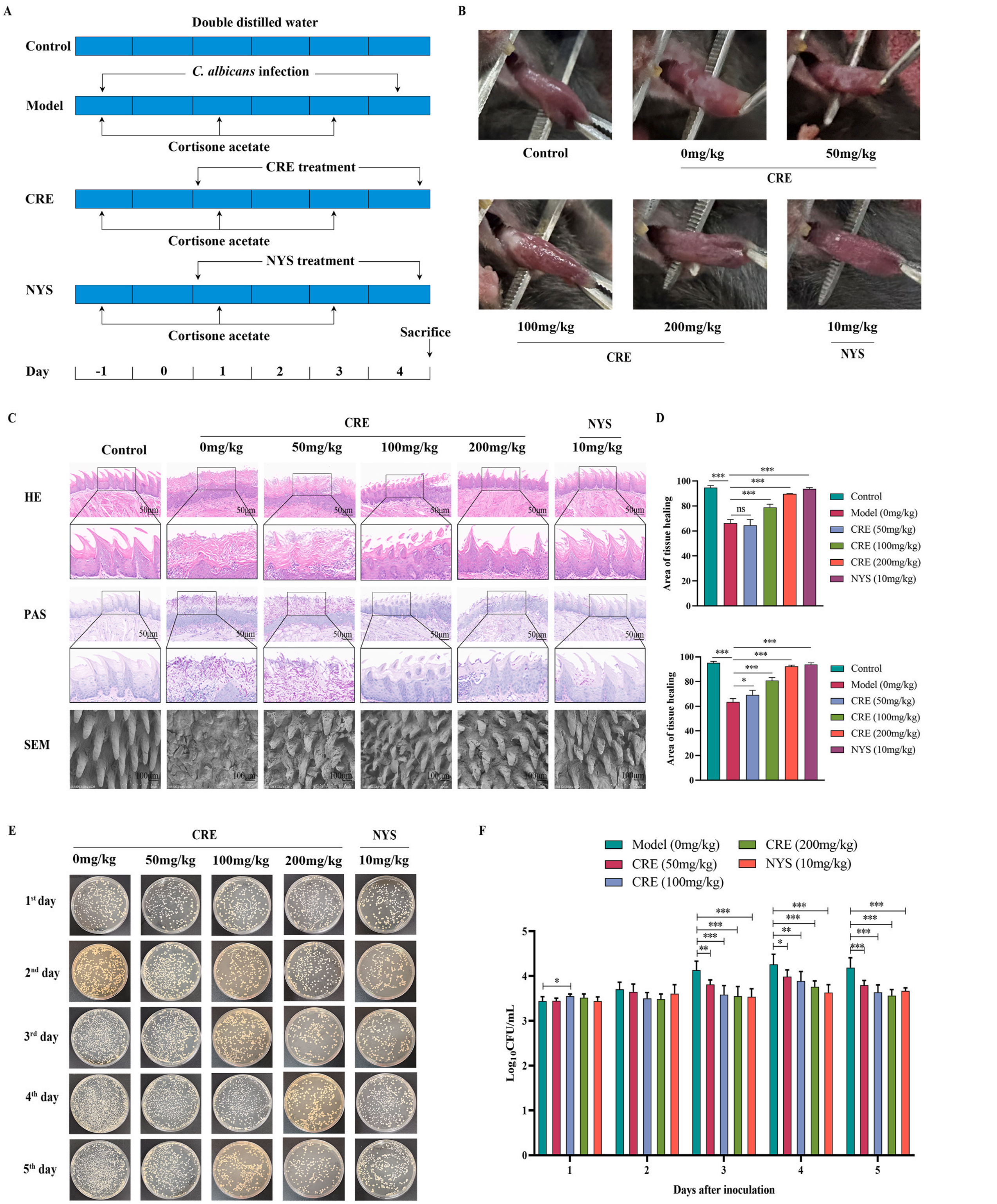


Fig. 2. Efficacy of CRE in improving OPC symptoms in a *Candida*-infected murine model. (A) The treatment procedure for the *Candida*-infected murine model with or without CRE or NYS. (B) Lesions on the mucosal surface of the tongue in mice. (C) H&E and PAS staining outcomes in the lesions were observed under × 200 (scale bar, 50 μm) and × 200 (scale bar, 50 μm) magnification. The bottom images represent SEM (× 500, scale bar: 100 μm) outcomes in the lesions of all groups. (D) The upper panel represents the quantitative analysis of the H&E staining of the tongue-filled tissue area, while the lower panel represents the quantitative analysis of the PAS staining of the tongue-filled tissue area. (E) *C. albicans* load count on the surface of the tongue tissue of mice treated with different doses of CRE or NYS. (F) The quantitative analysis of *C. albicans* load on the surface of the tongue tissue of mice treated with different doses of CRE or NYS. Results are the mean ± standard deviation. Asterisk represents significant, *p < 0.05, **p < 0.01, ***p < 0.001.

(SEM).

2.11. Transwell assay

The experimental procedure followed a previous study with some modifications (Gontijo et al., 2019). Approximately 3×10^5 cells were seeded in 24-well 0.4-μm Transwell inserts (Corning Inc.). Culture medium (0.5 mL and 1 mL, respectively) was added to the apical and basolateral compartments. *C. albicans* SC5314 was incubated on YPD agar medium at 37 °C for 24 h. Subsequently, some of the *C. albicans* colonies were inoculated on YPD liquid medium for 14 h. *C. albicans* was then seeded with FaDu cells at a concentration of 1×10^5 fungal cells/mL with or without BBR or NYS added to the apical chamber. The FaDu cells were cultured in a 37 °C, 5% CO_2 incubator for 24 h. The concentration of *C. albicans* (CFU/mL) in the basolateral compartment was then calculated after treatment with or without BBR or NYS.

2.12. Enzyme-linked immunosorbent assay (ELISA)

The levels of interleukin (IL)-1α, IL-1β, IL-6, granulocyte macrophage colony-stimulating factor (GM-CSF), and granulocyte-colony-stimulating factor (G-CSF) were assessed in all samples using ELISA kits. One portion of the sample comprised murine serum (serum obtained from sacrificed mice, centrifuged to collect the supernatant), while the other part consisted of cell supernatant (supernatant collected from treated FaDu cells after centrifugation). The enzyme-linked immunosorbent assays were conducted in accordance with the instructions provided by the manufacturer.

2.13. Immunofluorescence

Immunofluorescence analysis was conducted on tongue tissue and FaDu cells to examine the expression of gC1qR and EGFR proteins. First, 5-μm paraffin sections of tongue tissue were deparaffinized and blocked with 3% bovine serum albumin. The sections were then incubated overnight at 4 °C with EGFR antibody (1:3000). After incubation, horseradish peroxidase-labeled goat anti-rabbit secondary antibody (1:500) was added, followed by the addition of fluorescein isothiocyanate dye, microwaving, and incubation with gC1qR antibody (1:300) overnight at 4 °C. Finally, CY3-labeled goat anti-rabbit secondary antibody (1:300) was added. For FaDu cells, they were cultured in a 5% CO_2 cell culture chamber at 37 °C for 24 h, until the cell density reached approximately 80%. The *C. albicans*-stimulated FaDu cells were then treated with drug concentrations of 10 μM BBR and 25 μM NYS at a concentration of 1×10^5 CFU/ml. The cells were then fixed with 4% paraformaldehyde, permeabilized with 0.1% Triton X-100, blocked with ready-made normal goat serum, and washed with PBS. Secondary antibodies used were Alexa Fluor 488 goat anti-mouse (1:500) and Alexa Fluor 594 goat anti-rabbit (1:500). An anti-fluorescence quenching medium (containing DAPI) was added, and the samples were observed under a fluorescence microscope.

2.14. Transfection

Silencing RNA (siRNA) and a negative control (NC) targeting gC1qR were designed and synthesized by GenePharma (Shanghai, China). The siRNA sequences can be found in Supplementary Table 1. The plasmid vector gC1qR and pcDNA3.1 were obtained from GenePharma. Transfection was performed following the manufacturer's instructions, and four different groups (si-726, si-635, si-419, and si-871) were screened to identify the siRNA with the highest efficacy. The siRNA, NC, pcDNA3.1, and the gC1qR plasmid vector were transfected into FaDu cells according to the manufacturer's instructions. Cells were collected 24 h after transfection. The knockdown efficiency and the overexpression efficiency was evaluated using qRT-PCR and western blotting.

2.15. Western blotting

The solvent ratio for lysing tissues or cells consisted of RIPA buffer: protease inhibitor: phosphatase inhibitor in a ratio of 100:1:1. Total protein levels were determined using the BCA protein detection kit (Spark Jade, Shandong, China). The protein lysates were then separated by 10% sodium dodecyl sulfate-polyacrylamide gel electrophoresis and transferred onto polyvinylidene fluoride membranes. After blocking the membranes with 5% skim milk for 1–2 h, they were incubated with primary antibodies overnight at 4 °C, followed by a 1-h incubation with the secondary antibody. In western blotting, β-actin served as the loading control. Protein bands were visualized using an enhanced chemiluminescence detection reagent (Spark Jade, Shandong, China), and the grayscale analysis of the protein bands was carried out using ImageJ software.

2.16. qRT−PCR

Total RNA was isolated from treated tongue tissue and FaDu cells using the SPARKeasy Tissue/Cell Rapid Extraction Kit (Spark Jade, Shandong, China), and complementary DNA was synthesized using SPARKscript II. The RT Plus Reverse Transcription Kit (Spark Jade, Shandong, China) was employed for this purpose. Subsequently, qRT-PCR was carried out utilizing the qPCR SYBR Green Master Mix (Yeasen Biotechnology Co., Shanghai, China) according to the manufacturer's instructions. The primer sequences for β-actin, gC1qR, and EGFR were generated by Sangon (Sangon Biotech Co., Shanghai, China), and these sequences can be found in Supplementary Table 2 β-Actin mRNA levels were used as endogenous controls for the target genes. The fold change in expression was calculated as $2-^{\Delta\Delta Ct}$ ($\Delta\Delta Ct = \Delta Ct_{control} - \Delta Ct_{treatment}$).

2.17. Statistical analysis

The data were tested in triplicate, expressed as the mean ± standard deviation, and analyzed using SPSS 23.0 (SPSS Inc, Chicago, IL, USA). Graphs were generated using GraphPad Prism 9.5. For data with homogeneous variance, one-way analysis of variance (ANOVA) followed by LSD analysis was conducted. In the case of uneven variance, Games-Howell analysis was performed using the Welch ANOVA method. A p-value of less than 0.05 was considered statistically significant.

3. Results

3.1. 3.1. component analysis in CRE

As shown in Fig. 1, we evaluated the chemical components of CRE. Fig. 1A displays the natural raw materials of *C. rhizoma*. Utilizing UPLC-Q/TOF-MS^E, we investigate the active ingredients within CRE, as depicted in Fig. 1B. Through component analysis of CRE and cross-referencing the MS database, we have identified a minimum of 18 distinct chemical compounds, as shown in Fig. 1C.

3.2. Effect of CRE on morphological changes and fungal load in a Candida-infected murine model

To investigate the therapeutic efficacy of CRE against oral mucosal infections, an experimental murine model with *Candida* inoculation was formulated, and a specific dosing protocol was devised for the study. The therapeutic intervention extended over a four-day period, as depicted in Fig. 2A. Observations revealed that the control group's mouse tongues were of normal appearance - smooth and moist with an intact mucosal surface. In contrast, mice not treated with CRE developed white, pseudomembrane-covered tongues with severe mucosal damage. Increasing CRE dosages progressively reduced the pseudomembrane on the tongue's surface, leaving fewer white spots and markedly restoring

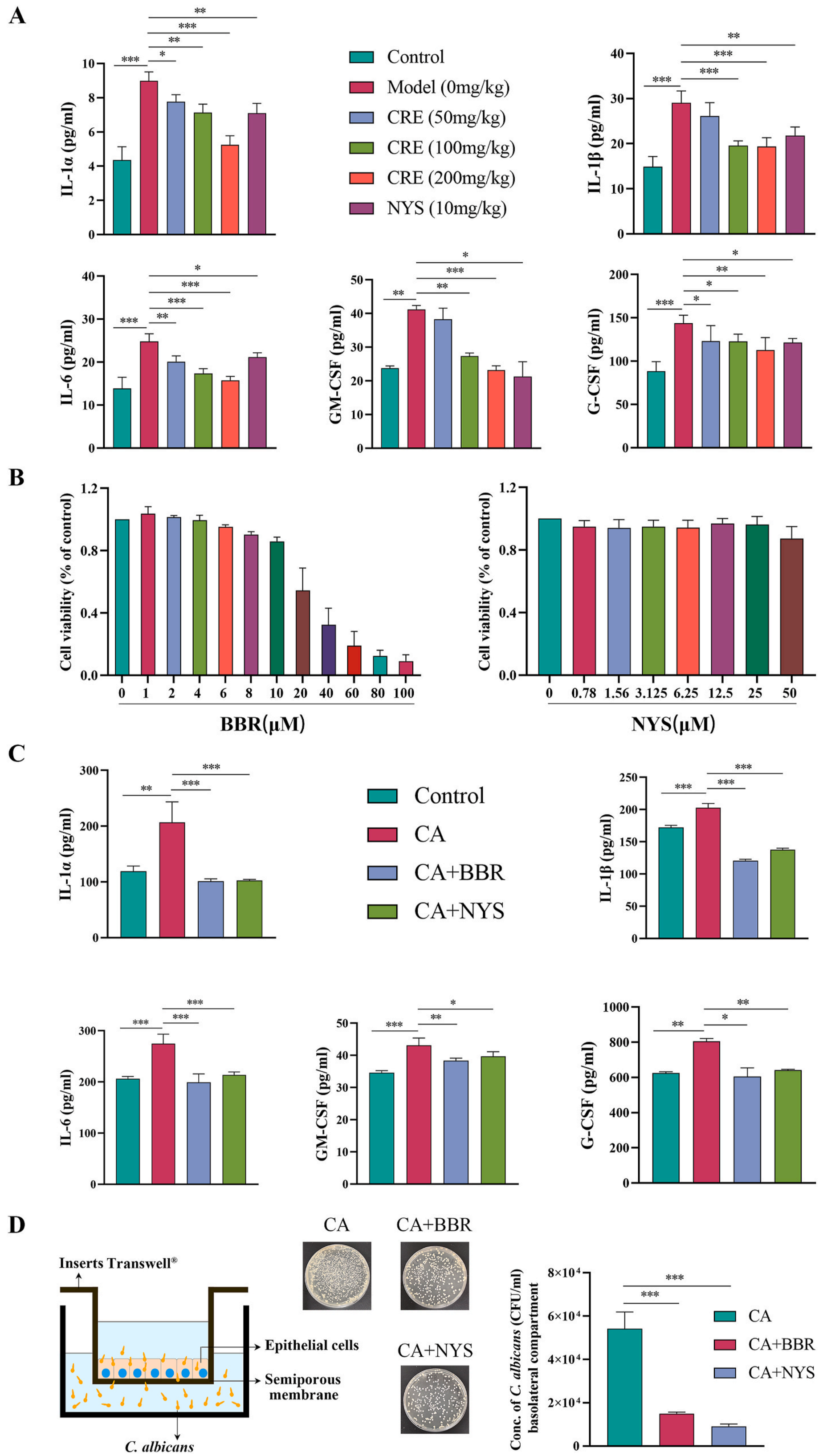


Fig. 3. Inflammatory factor levels in serum of *Candida*-infected murine model and FaDu cells, and invasive ability assessment of *C. albicans* to epithelial cells. (A) On the fifth day, the mice treated with or without CRE or NYS were sacrificed to collect the serum and various tissues. The levels of IL-1α, IL-β, IL-6, GM-CSF, and G-CSF in the serum were detected using ELISA. (B) The left panel represents the cell viability of FaDu cells treated with different concentrations of BBR, and the right panel represents the cell viability of FaDu cells treated with different concentrations of NYS. (C) *Candida*-infected FaDu cells treated with or without BBR or NYS; the supernatants were collected to examine IL-1α, IL-β, IL-6, GM-CSF, and G-CSF contents using ELISA. (D) Concentration of *C. albicans* (CFU/mL) in the basolateral compartment of the Transwell apparatus after 24-h treatment with BBR or NYS. Results represent the mean ± standard deviation. Asterisk indicates significance; *p < 0.05, **p < 0.01, ***p < 0.001.

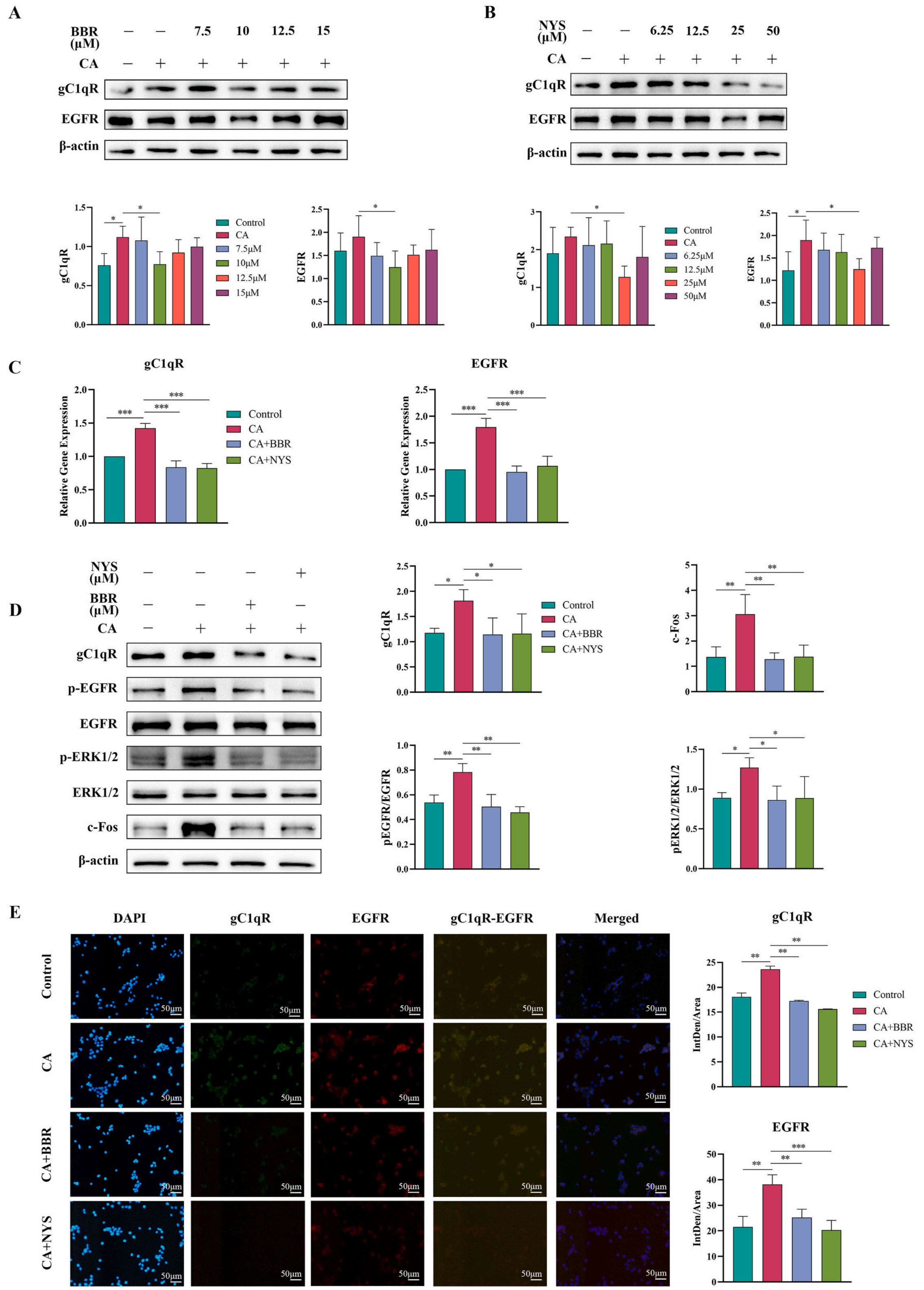


Fig. 4. Mechanism and functional characterization of FaDu cells treated with BBR or NYS under the stimulation of *C. albicans*. (A) The upper panel represents the optimal concentration of western blotting-screened BBR on *C. albicans*-stimulated FaDu cells, while the lower panel represents the quantitative analysis of gC1qR and EGFR proteins. (B) The upper panel represents the optimal concentration of western blotting-screened NYS for *C. albicans*-stimulated FaDu cells, and the lower panel represents the quantitative analysis of gC1qR and EGFR proteins. (C) *Candida*-infected FaDu cells treated with or without BBR or NYS, RNA was extracted, and the mRNA expression of gC1qR and EGFR was detected using qRT-PCR. (D) The left panel represents the protein expression of gC1qR, p-EGFR, EGFR, p-ERK1/2, ERK1/2, and c-Fos in *Candida*-infected FaDu cells treated with or without BBR or NYS, and the right panel represents the quantitative analysis of western blotting. (E) The left panel represents the expression of gC1qR and EGFR colocalization in *Candida*-infected FaDu cells treated with or without BBR or NYS, and the right panel represents the quantitative analysis of gC1qR and EGFR immunofluorescence.

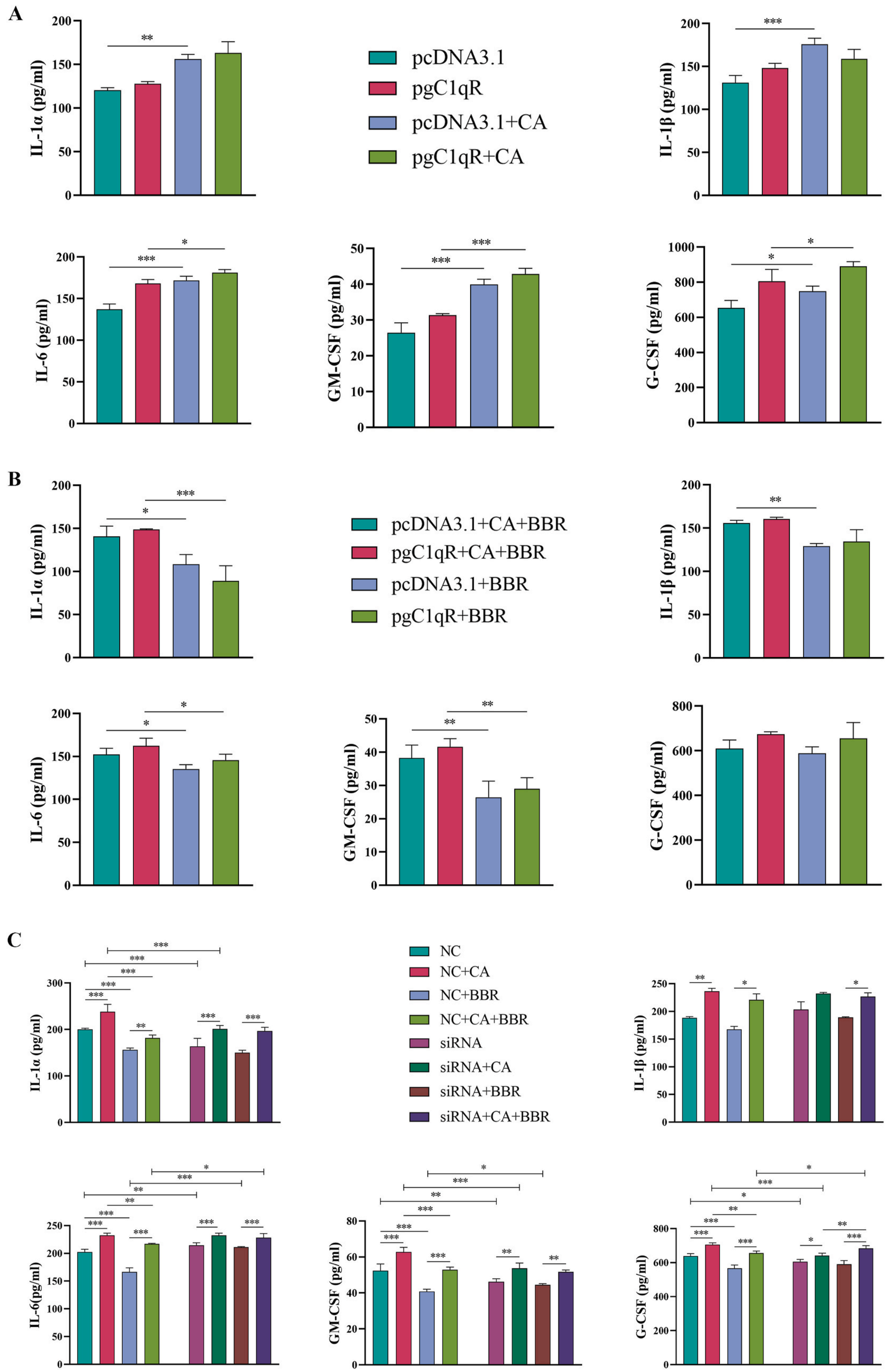


Fig. 5. Overexpression or silencing of gC1qR receptors, inflammatory factor levels in the serum of *Candida*-infected or uninfected FaDu cells were observed after treatment with or without BBR. (A) The levels of IL-1α, IL-β, IL-6, GM-CSF, and G-CSF in the serum were detected using ELISA after transfection of pcDNA3.1, and pgC1qR in *Candida*-infected or uninfected FaDu cells. (B) IL-1α, IL-β, IL-6, GM-CSF, and G-CSF levels in the supernatant of *Candida*-infected or uninfected FaDu cells were treated with BBR or without BBR using ELISA. (C) Silencing gC1qR, *Candida*-infected or uninfected, with or without BBR treatment, the supernatants were collected to examine IL-1α, IL-β, IL-6, GM-CSF, and G-CSF contents using ELISA.

the tongue's integrity (Fig. 2B). Further examination to assess the impact of CRE on tongue mucosal damage in Candida-infected mice indicated that those without CRE treatment exhibited significant dorsal papilla damage and heavy infiltration of inflammatory cells concurrent with profuse hyphal growth. Higher CRE doses corresponded to less lingual mucosal structure damage and a pronounced reduction in hyphal penetration and inflammatory cell invasion, suggesting a preservation of the tongue's structural integrity (Fig. 2C). Scanning Electron Microscopy (SEM) confirmed that tongues from mice not receiving CRE presented withered filiform papillae and extensive mucosal damage. In contrast, increasing CRE concentrations led to orderly filiform papillae and gradual mucosal healing (Fig. 2C). Quantification of H&E and PAS-stained damaged areas (Ulian-Benitez et al., 2022) allowed the observation of mucosal repair at varying CRE concentrations (Fig. 2D). Unlike control mice, those in the non-CRE group needed larger repairs of tongue tissue. This requirement diminished as CRE dosage was enhanced (Fig. 2D). Fungal burden assessments in the mice's oral cavity revealed significant reductions with higher CRE dosages, in contrast to the non-CRE group (Fig. 2E). Additionally, as CRE treatment continued, therapeutic effects became more notable and dose-dependent. The high-dose CRE group showed similar therapeutic outcomes to the group treated with Nystatin (NYS). CRE also notably reduced fungal burdens in several other organs (liver, lungs, stomach, spleen, kidneys, colon) in a dose-responsive manner post-treatment. Furthermore, CRE positively adjusted the weight of the mice in a dose-dependent pattern (Supplementary Fig. 1). Consequently, CRE significantly diminished the *C. albicans* invasion, leading to an improvement in symptoms of OPC.

3.3. The effect of CRE and its main component BBR on the levels of inflammatory factors in Candida-infected murine model and supernatant of FaDu cells

Upon *C. albicans* infection, activation of EGFR in host epithelial cells fostered anti-*C. albicans* endocytosis, prompting secretion of pro-inflammatory cytokines and chemokines, orchestrating the host defense. Serological ELISA in mice revealed marked elevation of inflammatory markers like IL-1α, IL-1β, and IL-6, along with colony-stimulatory factors (GM-CSF and G-CSF) in the untreated group, contrasting with the dose-responsive reduction post-CRE administration (Fig. 3A). Equivalent levels to NYS were noted in the high-dose CRE cohort for IL-1β, GM-CSF, and G-CSF. Disparate minimum inhibitory concentrations of CRE on FaDu cells and *C. albicans* resulted in notably diminished FaDu cell viability post-CRE treatment in *C. albicans*-stimulated environments (data not shown). BBR, a key CRE constituent, exhibited *C. albicans* suppression (Zorić et al., 2017). Subsequent investigations probed BBR's influence on *C. albicans*-exposed FaDu cells, with survival rates at 1×10^5 CFU/mL between 70% and 80% (Supplementary Fig. 2). The impact of BBR and NYS on viability subsequent to administration was quantified, noting ~80% viability with 10 μM BBR and nominal impact with 25 μM NYS (Fig. 3B). Upon examining IL-1α, IL-1β, IL-6, GM-CSF, and G-CSF in FaDu supernatants via ELISA, a rise in inflammatory factor levels post-*C. albicans* intervention was mitigated following BBR or NYS treatment, underlining BBR's efficacy against *C. albicans*-induced inflammatory response (Fig. 3C). Additionally, *C. albicans*' aptitude for breaching the FaDu epithelial fusion barrier was assessed using a Transwell assay to gauge BBR's barrier-preserving effects. Without treatment, *C. albicans* demonstrated robust epithelial penetration, yet BBR or NYS therapy fortified FaDu cell barrier function, inhibited *C. albicans* infiltration, and notably lessened basolateral CFU counts (Fig. 3D). These findings imply that CRE and its main component, BBR, lessen inflammatory mediators and BBR hinders *C. albicans* invasion by enhancing FaDu cell barrier fusion capability.

3.4. 3.4. BBR regulates the gC1qR-EGFR/ERK/c-Fos signaling pathway by influencing the gC1qR-EGFR co-receptor

BBR exhibits an inhibitory effect on *C. albicans*, nevertheless, the optimal concentration for stimulating FaDu cells with *C. albicans* requires further validation. Based on previous studies by other researchers (Kuo et al., 2005; Seo et al., 2015), we conducted additional screening to determine the optimal BBR concentration using western blotting. The findings indicated that at 10 μM, BBR had a more significant impact on FaDu cells under *C. albicans* stimulation (Fig. 4A). Based on previous studies by other researchers (Boros-Majewska et al., 2015), western blotting was employed to further identify the optimal NYS drug concentration. The results demonstrated that at 25 μM, NYS exhibited a more significant effect on FaDu cells stimulated by *C. albicans* (Fig. 4B). Alterations in gC1qR and EGFR mRNA levels were observed in FaDu cells. It was observed that under *C. albicans* stimulation, the mRNA transcription of gC1qR and EGFR increased. Following treatment with BBR or NYS, the mRNA transcription decreased (Fig. 4C).

Changes in gC1qR, p-EGFR/EGFR, p-ERK1/2/ERK1/2, and c-Fos expression in FaDu cells were analyzed. The results suggested that protein expression significantly increased in *C. albicans*-stimulated FaDu cells. After treatment with BBR or NYS, protein expression was significantly reduced (Fig. 4D). Quantitative analyses of gC1qR, p-EGFR/EGFR, p-ERK1/2/ERK1/2, and c-Fos proteins are presented on the right side of Fig. 4D. The colocalization expression of gC1qR and EGFR was examined using immunofluorescence. The results indicated that fluorescence intensities of gC1qR and EGFR increased in FaDu cells stimulated with *C. albicans*. Following treatment with BBR or NYS, the fluorescence intensities of gC1qR and EGFR decreased, and there was almost no gC1qR fluorescence in the NYS treatment group (Fig. 4E). The right side of Fig. 4E presents quantitative immunofluorescence analysis of gC1qR and EGFR. These findings suggest that BBR enhances the endocytosis of FaDu cells against *C. albicans* by impacting the gC1qR EGFR/ERK/c-Fos signaling pathway, thereby inhibiting the expression of related genes and proteins.

3.5. Improvement of inflammatory cytokine levels in FaDu cells by verifying the effect of BBR on gC1qR receptors through gene overexpression and silencing

To further confirm the impact of BBR on the endocytosis of FaDu cells against *C. albicans* through modulation of the gC1qR receptor, overexpression and silencing of the gC1qR receptor were performed to observe the expression levels of inflammatory factors. Utilizing pcDNA3.1 as a control, the gC1qR plasmid was transfected into FaDu cells. Successful transfection of the gC1qR plasmid was confirmed by qRT-PCR and Western blotting under *C. albicans* stimulation conditions. Subsequent qRT-PCR and Western blotting analyses revealed a substantial increase in the expression of the gC1qR group, signifying successful plasmid transfection (Supplementary Fig. 3). The FaDu cells supernatant was collected, and the levels of IL-1α, IL-1β, IL-6, GM-CSF, and G-CSF inflammatory cytokines were measured using ELISA. Upon *C. albicans* stimulation, the control group exhibited elevated levels of IL-1α, IL-1β, IL-6, GM-CSF, and G-CSF inflammatory cytokines. Following *C. albicans* stimulation in the gC1qR group, the expression of IL-1α and IL-1β inflammatory factors showed no significant difference, while IL-6, GM-CSF, and G-CSF inflammatory factors did not exhibit substantial expression (Fig. 5A). In Fig. 5B, following the findings of Fig. 5A, a drug replenishment experiment was conducted. The results demonstrated decreased levels of IL-1α, IL-1β, IL-6, and GM-CSF inflammatory factors after BBR intervention, with no significant presence of G-CSF (Fig. 5B). The siRNA sequence with the most effective silencing effect was identified among four siRNA sequences: si-726, si-635, si-419, and si-871. The results from qRT-PCR and Western blotting analyses indicated that si-726 achieved the best silencing effect (Supplementary Fig. 4). Subsequent experiments employing si-726, collection of FaDu cell

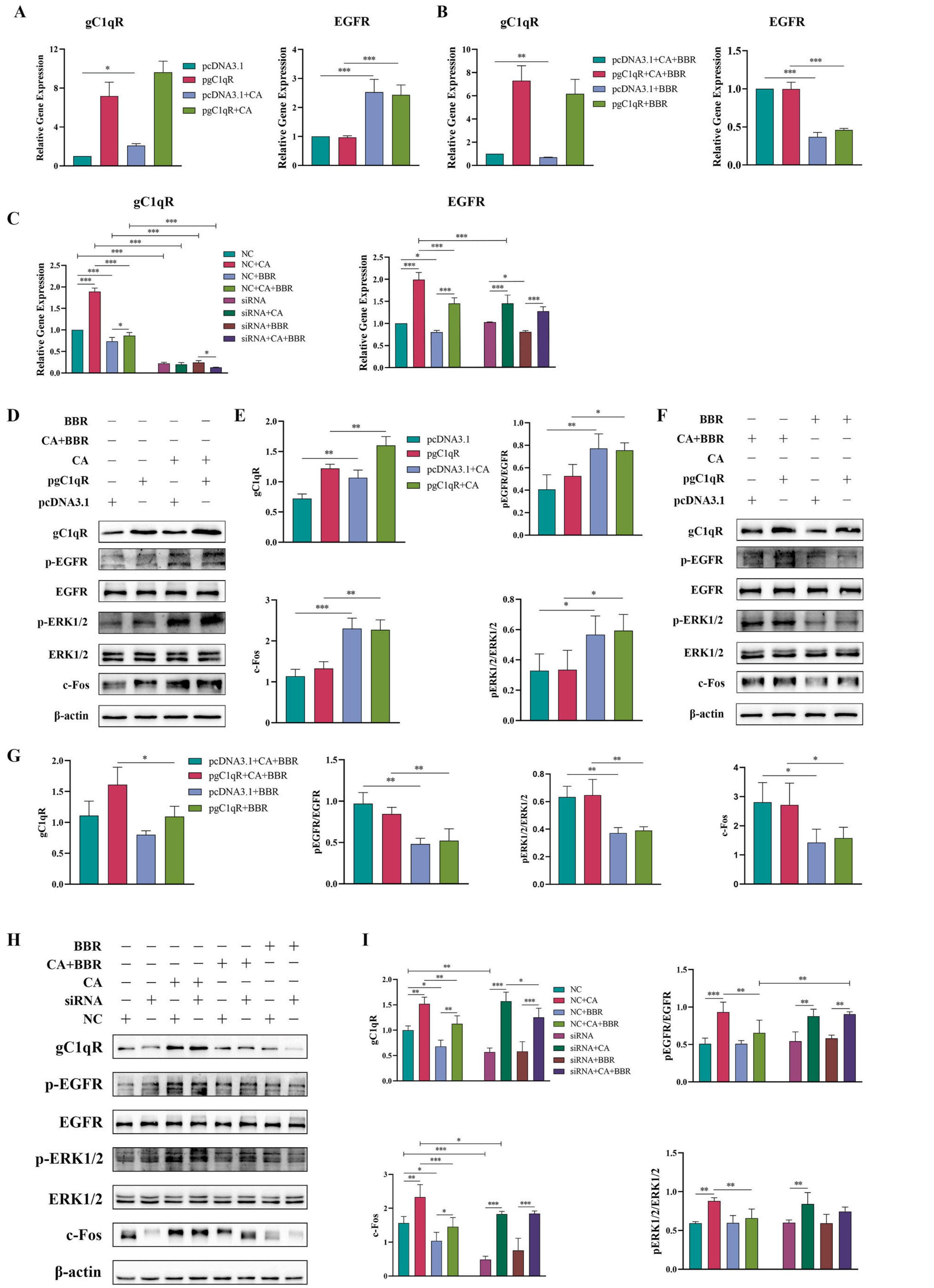


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Fig. 6. Overexpression or silencing of gC1qR receptors, mechanism of *Candida*-infected or uninfected FaDu cells after treatment with or without BBR. (A) Overexpression of gC1qR, *Candida*-infected or uninfected FaDu cells, RNA was extracted, and the mRNA expression of gC1qR and EGFR was detected using qRT-PCR. (B) Based on (A), treatment with or without BBR, RNA was extracted, and the mRNA expression of gC1qR and EGFR was detected using qRT-PCR. (C) Silencing gC1qR, *Candida*-infected or uninfected FaDu cells were treated with or without BBR, RNA was extracted, and the mRNA expression of gC1qR and EGFR was detected using qRT-PCR. (D) After gC1qR overexpression and *Candida*-infected or uninfected FaDu cells, the protein expression of gC1qR, p-EGFR, EGFR, p-ERK1/2, ERK1/2, and c-Fos was detected using western blotting. (E) Quantitative analysis of (D) protein. (F) Based on (D), treatment with BBR or without BBR, and the protein expression of gC1qR, p-EGFR, EGFR, p-ERK1/2, ERK1/2, and c-Fos was detected using western blotting. (G) The quantitative analysis of (F) protein. (H) Silencing gC1qR, *Candida*-infected or uninfected FaDu cells were treated with BBR or without BBR, and the protein expression of gC1qR, p-EGFR, EGFR, p-ERK1/2, ERK1/2, and c-Fos was detected using western blotting. (I) The quantitative analysis of (H) protein.

supernatant, and ELISA detection of IL-1α, IL-1β, IL-6, GM-CSF, and G-CSF inflammatory cytokine levels revealed an increase in the levels of inflammatory factors upon *C. albicans* stimulation. However, in the absence of *C. albicans* stimulation, BBR treatment was able to reduce the levels of inflammatory factors (Fig. 5C). These results suggest that BBR can influence the gC1qR receptor to enhance the endocytosis of *C. albicans* by FaDu cells, thereby reducing the expression levels of inflammatory factors.

3.6. Confirming the effects of BBR on gC1qR receptors by gene overexpression and silencing, and on the gC1qR-EGFR/ERK/c-Fos signaling pathway

BBR's effects on the gC1qR was confirmed through qRT-PCR and western blotting. Unlike the pcDNA3.1 control group, overexpression of the gC1qR gene was significantly observed in the gC1qR plasmid transfection group, while the EGFR gene remained unaffected. Furthermore, the gene expression of gC1qR and EGFR increased following *C. albicans* stimulation (Fig. 6A). Subsequently, BBR treatment resulted in varying degrees of reduction in gC1qR and EGFR gene expression (Fig. 6B). The results of qRT-PCR silencing demonstrated that, in contrast to the NC group, stimulation with *C. albicans* notably increased the demonstration of gC1qR and EGFR genes in the NC + CA group. Moreover, the gene expression of gC1qR decreased significantly in the siRNA group, whereas EGFR gene expression exhibited little effect (Fig. 6C). In addition, the NC + BBR group showed lower levels of gC1qR and EGFR compared to the NC + CA + BBR group (Fig. 6C). These results validate BBR's effects on the gC1qR receptor and its influence on gene expression in FaDu cells. Furthermore, unlike the pcDNA3.1 control group, substantial addition of gC1qR protein was observed in the transfected gC1qR plasmid group, with minimal effect on the expression of other related proteins. Infection with *C. albicans* led to a significant increase in the protein levels of gC1qR, p-EGFR/EGFR, p-ERK1/2/ERK1/2, and c-Fos (Fig. 6D and E). However, these protein levels diminished after BBR treatment (Fig. 6F and G). Western blot silencing results indicated that the siRNA group exhibited decreased levels of gC1qR and c-Fos proteins, while other related proteins remained largely unchanged, as compared to the NC group. Additionally, the proteins gC1qR and c-Fos decreased after BBR therapy, differing from the protein levels following *C. albicans* stimulation (Fig. 6H and I). These findings demonstrate that BBR regulates the gC1qR-EGFR/ERK/c-Fos signaling pathway through its effect on the gC1qR receptor from both genetic and protein expression perspectives.

3.7. CRE regulates the gC1qR-EGFR/ERK/c-Fos signaling pathway by influencing the gC1qR-EGFR co-receptor

Subsequently, we examined the impact of CRE on the gC1qR-EGFR co-receptor in an in vivo setting. The qRT-PCR results revealed a significant increase in the expression of gC1qR and EGFR genes in the model group without CRE treatment, which gradually decreased in a dose-dependent manner following CRE administration; notably, the high-dose CRE group exhibited a treatment effect similar to NYS (Fig. 7A). Analysis through western blotting demonstrated elevated protein expression levels of gC1qR, p-EGFR/EGFR, p-ERK1/2/ERK1/2, and c-Fos in the absence of CRE treatment; however, these protein levels gradually declined with dose-dependent CRE treatment, eventually approaching levels seen with NYS treatment (Fig. 7B). Subsequent assessment of the gC1qR-EGFR co-receptor expression using immunofluorescence revealed significantly increased fluorescence intensities of gC1qR and EGFR in the model group without CRE treatment compared to the control group. Interestingly, the filamentous papillae on the tongue mucosa surface showed severe damage in this group, but following dose-dependent CRE treatment, the fluorescence intensity of gC1qR and EGFR decreased gradually, along with regeneration of the filamentous papillae on the tongue mucosa, appearing partially intact (Fig. 7C). These findings indicate that CRE has an influence on the gC1qR-EGFR co-receptor and plays a regulatory role in the gC1qR-EGFR/ERK/c-Fos signaling pathway. This investigation proposes a promising new therapeutic approach for oral candidiasis treatment.

4. Discussion

TCM is widely used in the treatment of oral diseases. Hitomi et al. (2016) demonstrated that a traditional Japanese medicine known as Hangshashito, derived from extracts of seven herbs, possesses anti-inflammatory properties, offering relief from oral mucositis and reducing the discomfort associated with mouth ulcers. Yang et al. (2016) unveiled that Gan-Lu-Yin (GLY), a TCM remedy, exhibits efficacy in the clinical treatment of ulcerative stomatitis and also shows promise in enhancing oral cancer therapy by reducing TNF-α cytokine levels. Additionally, Wang et al. (2018) reported that the TCM Chining decoction (CHIN) exhibits notable therapeutic benefits in managing radiation-induced oral mucositis in patients with head and neck cancer. Our study outcomes indicate that CRE exhibits favorable therapeutic efficacy in addressing oral candidiasis while also aiding in the restoration of damaged mucosal surfaces of the tongue.

C. albicans plays a pivotal role in mucosal infections by breaching the mucous membrane barrier via hyphae. The primary surface invasion factors of *C. albicans*, extensively studied and recognized, include the Als3 and Ssa1 proteins (Höfs et al., 2016; Lu et al., 2014). Among these factors, EGFR plays a crucial role in initiating endocytosis in epithelial cells and provoking pro-inflammatory reactions. Activation of the downstream clathrin-dependent pathway occurs upon ligand-receptor binding, triggering endocytosis (Liu and Filler, 2011; Moreno-Ruiz et al., 2009). Secretion of Clys by *C. albicans* hyphae induces the influx of calcium ions into epithelial cells, activating matrix metalloproteinases, causing the shedding of native EGFR ligands, consequently activating EGFR (Ponde et al., 2022). During the infection of epithelial cells by *C. albicans*, hyphal-secreted Clys activates EGFR in epithelial cells, prompting mitogen-activated protein kinase (MAPK) signaling responses (Ponde et al., 2022). This study have verified a significant increase in inflammatory factors, as well as the upregulation of related genes and proteins, following stimulation by *C. albicans* at both animal and cell levels. Acting as a critical co-receptor alongside EGFR, gC1qR has been identified as a vital co-factor for EGFR during *C. albicans* invasion, promoting enhanced EGFR expression, elevated inflammation levels, and exacerbating *C. albicans* infections (Phan et al., 2021). The findings of this research reveal that CRE can diminish the expression of the gC1qR-EGFR co-receptor in mouse tongue tissues, thereby down-regulating the EGFR/ERK/c-Fos signaling pathway and subsequently reducing the expression of inflammatory factors, target genes, and

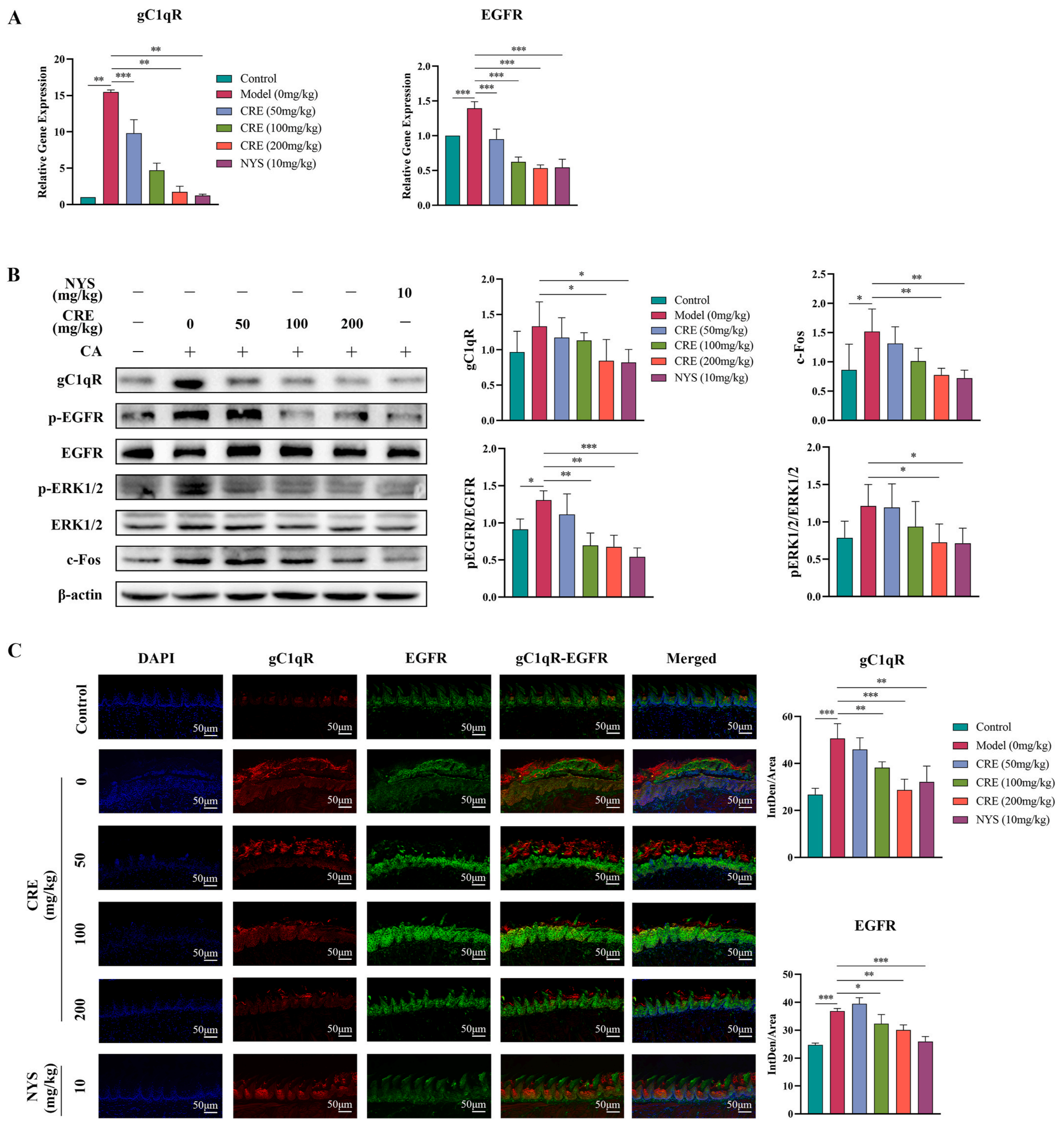

Fig. 7. Mechanism of CRE in a *Candida*-infected murine model of ameliorating *Candida* infection. (A) In the Candida-infected murine model treated with or without CRE or NYS, RNA was extracted, and qRT-PCR was used to detect the mRNA expression of gC1qR and EGFR. (B) The left panel represents the *Candida*-infected murine model with or without CRE or NYS, and the protein expression of gC1qR, p-EGFR, EGFR, p-ERK1/2, ERK1/2, and c-Fos was detected using western blotting, and the right panel represents the quantitative analysis of protein. (C) Immunofluorescence colocalizes the expression of gC1qR and EGFR in the *Candida*-infected murine model with or without CRE or NYS, and the right panel represents the quantification of immunofluorescence.

associated proteins. In the context of the EGFR-mediated immune response against fungal infections, research findings indicate that the primary pathway activated by the EGFR adaptor is ERK1/2, rather than p38 or SAPK/JNK. This pathway is responsible for c-Fos activation through the MAPK-ERK1/2 signaling cascade (Ponde et al., 2022). Subsequently, this cascade triggers the production of IL-1α, IL-1β, IL-6, GM-CSF, and G-CSF, leading to the initiation of the host immune response. The outcomes of this study demonstrate that BBR can diminish the expression of the gC1qR-EGFR co-receptor on FaDu cells upon stimulation by *C. albicans*. This downregulation impacts the EGFR/ERK/c-Fos signaling pathway, thereby reducing the expression of inflammatory factors, target genes, and associated proteins.

This study demonstrated the therapeutic benefits of CRE in the OPC model and BBR in *C. albicans*-infected FaDu cells, shedding light on the gC1qR-EGFR co-receptor and associated signaling pathways. It provided initial insights into the mechanism of action of CRE and its primary component BBR in treating OPC. Traditional antifungal medications (e.g., fluconazole) for OPC primarily target the structure or composition of *C. albicans*. However, this research focused on how CRE or BBR influences host immunity by affecting the gC1qR-EGFR co-receptor. Furthermore, through manipulating the expression of gC1qR, this study revealed that BBR can modulate EGFR expression by impacting gC1qR, thereby influencing the EGFR/ERK/c-Fos signaling pathway and dampening the immune response. These findings offer a novel approach to OPC treatment and pave the way for potential new directions in OPC therapy through experimental exploration.

This study focused on the potential of CRE or BBR to influence the gC1qR-EGFR co-receptor for therapeutic effects on OPC. Further research is necessary to confirm whether BBR can indeed impact the EGFR receptor by targeting the gC1qR receptor through molecular docking or molecular dynamics simulation. Furthermore, our findings indicated that the rational silencing of the gC1qR receptor effectively reduced EGFR receptor stimulation, leading to decreased secretion of inflammatory cytokines. However, experimental results revealed an increase in the expression of inflammatory cytokines IL-1β and IL-6, a similar situation occurred in the study of Phan et al. (2021), but no explanation was given. Therefore, we aim to conduct further investigations to investigate whether CRE and its primary component BBR can target the gC1qR receptor to regulate the EGFR-MAPK signaling pathway, providing a comprehensive evaluation of the mechanism of OPC treatment.

5. Conclusion

After observing the stimulation of the gC1qR-EGFR co-receptor by *C. albicans*, we explored the potential of CRE and its primary component BBR to influence the downstream MAPK signaling pathway through modulation of the gC1qR-EGFR co-receptor and orchestrate the host immune response. This experimental investigation confirmed these effects through the assessment of inflammatory factors and their related target genes and proteins, accompanied by the examination of gC1qR-EGFR co-receptor localization using immunofluorescence. Ultimately, it was established that CRE and its primary component BBR can impact the gC1qR-EGFR co-receptor, presenting a novel and effective strategy for the treatment of OPC.

CRediT authorship contribution statement

Mengyuan Bao: Writing – review & editing, Writing – original draft, Investigation. **Qingru Bu:** Visualization, Validation, Investigation. **Min Pan:** Software. **Ran Xu:** Supervision. **Yujie Chen:** Validation. **Yue Yang:** Visualization, Validation. **Changzhong Wang:** Resources. **Tianming Wang:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2024.118305.

References

Authier, H., Bardot, V., Berthomier, L., Bertrand, B., Blondeau, C., Holowacz, S., Coste, A., 2022. Synergistic effects of licorice root and walnut leaf extracts on gastrointestinal candidiasis, inflammation and Gut microbiota composition in mice. Microbiol. Spectr. 10 (2), e0235521 https://doi.org/10.1128/spectrum.02355-21.

Boros-Majewska, J., Turczyk, Ł., Wei, X., Milewski, S., Williams, D.W., 2015. A novel in vitro assay for assessing efficacy and toxicity of antifungals using human leukaemic cells infected with Candida albicans. J. Appl. Microbiol. 119 (1), 177–187. https://doi.org/10.1111/jam.12817.

Chen, M., Cheng, T., Xu, C., Pan, M., Wu, J., Wang, T., Wu, D., Yan, G., Wang, C., Shao, J., 2022. Sodium houttuyfonate enhances the mono-therapy of fluconazole on oropharyngeal candidiasis (OPC) through HIF-1α/IL-17 axis by inhibiting cAMP mediated filamentation in Candida albicans-Candida glabrata dual biofilms. Virulence 13 (1), 428–443. https://doi.org/10.1080/21505594.2022.2035066.

Conti, H.R., Huppler, A.R., Whibley, N., Gaffen, S.L., 2014. Animal models for candidiasis. Curr. Protoc. Im. 105, 19.16.11–19.16.17. https://doi.org/10.1002/0471142735.im1906s105.

Geng, Q.S., Liu, R.J., Shen, Z.B., Wei, Q., Zheng, Y.Y., Jia, L.Q., Wang, L.H., Li, L.F., Li, J., Xue, W.H., 2021. Transcriptome sequencing and metabolome analysis reveal the mechanism of Shuanghua Baihe Tablet in the treatment of oral mucositis. Chin. J. Nat. Med. 19 (12), 930–943. https://doi.org/10.1016/s1875-5364(22)60150-x.

Gontijo, A.V., Ad, G.S., Koga-Ito, C.Y., Salvador, M.J., 2019. Biopharmaceutical and antifungal properties of ellagic acid-cyclodextrin using an in vitro model of invasive candidiasis. Future Microbiol. 14, 957–967. https://doi.org/10.2217/fmb-2019-0107.

Hao, Y., Huo, J., Wang, T., Sun, G., Wang, W., 2020. Chemical profiling of Coptis rootlet and screening of its bioactive compounds in inhibiting Staphylococcus aureus by UPLC-Q-TOF/MS. J. Pharm. Biomed. Anal. 180, 113089 https://doi.org/10.1016/j.jpba.2019.113089.

Hitomi, S., Ono, K., Yamaguchi, K., Terawaki, K., Imai, R., Kubota, K., Omiya, Y., Hattori, T., Kase, Y., Inenaga, K., 2016. The traditional Japanese medicine hangeshashinto alleviates oral ulcer-induced pain in a rat model. Arch. Oral Biol. 66, 30–37. https://doi.org/10.1016/j.archoralbio.2016.02.002.

Höfs, S., Mogavero, S., Hube, B., 2016. Interaction of Candida albicans with host cells: virulence factors, host defense, escape strategies, and the microbiota. J. Microbiol. 54 (3), 149–169. https://doi.org/10.1007/s12275-016-5514-0.

Hu, K., Zhang, H., Shi, G., Wang, B., Wu, D., Shao, J., Wang, T., Wang, C., 2023. Effects of n-butanol extract of Pulsatilla decoction on the NLRP3 inflammasome in macrophages infected with Candida albicans. J. Ethnopharmacol. 304, 116041 https://doi.org/10.1016/j.jep.2022.116041.

Hu, L., He, C., Zhao, C., Chen, X., Hua, H., Yan, Z., 2019. Characterization of oral candidiasis and the Candida species profile in patients with oral mucosal diseases. Microb. Pathog. 134, 103575 https://doi.org/10.1016/j.micpath.2019.103575.

Jiang, X.W., Zhang, Y., Zhu, Y.L., Zhang, H., Lu, K., Li, F.F., Peng, H.Y., 2013. Effects of berberine gelatin on recurrent aphthous stomatitis: a randomized, placebo-controlled, double-blind trial in a Chinese cohort. Oral Surg Oral Med Oral Pathol Oral Radiol 115 (2), 212–217. https://doi.org/10.1016/j.oooo.2012.09.009.

Kuo, C.L., Chi, C.W., Liu, T.Y., 2005. Modulation of apoptosis by berberine through inhibition of cyclooxygenase-2 and Mcl-1 expression in oral cancer cells. In Vivo 19 (1), 247–252.

Liu, Y., Filler, S.G., 2011. Candida albicans Als3, a multifunctional adhesin and invasin. Eukaryot. Cell 10 (2), 168–173. https://doi.org/10.1128/ec.00279-10.

Lu, Y., Su, C., Liu, H., 2014. Candida albicans hyphal initiation and elongation. Trends Microbiol. 22 (12), 707–714. https://doi.org/10.1016/j.tim.2014.09.001.

Moreno-Ruiz, E., Galán-Díez, M., Zhu, W., Fernández-Ruiz, E., d'Enfert, C., Filler, S.G., Cossart, P., Veiga, E., 2009. Candida albicans internalization by host cells is mediated by a clathrin-dependent mechanism. Cell Microbiol. 11 (8), 1179–1189. https://doi.org/10.1111/j.1462-5822.2009.01319.x.

Moyes, D.L., Wilson, D., Richardson, J.P., Mogavero, S., Tang, S.X., Wernecke, J., Höfs, S., Gratacap, R.L., Robbins, J., Runglall, M., Murciano, C., Blagojevic, M., Thavaraj, S., Förster, T.M., Hebecker, B., Kasper, L., Vizcay, G., Iancu, S.I., Kichik, N., Häder, A., Kurzai, O., Luo, T., Krüger, T., Kniemeyer, O., Cota, E., Bader, O., Wheeler, R.T., Gutsmann, T., Hube, B., Naglik, J.R., 2016. Candidalysin is a fungal peptide toxin critical for mucosal infection. Nature 532 (7597), 64–68. https://doi.org/10.1038/nature17625.

Niu, H.X., Duan, Q.J., Shi, G.X., Wu, D.Q., Shao, J., Wang, T.M., Wang, C.Z., 2019. Inhibitory effect of extract of Coptidis Rhizoma on invasion of Candida albicans hyphae in vitro. Zhongguo Zhong Yao Za Zhi 44 (1), 125–130. https://doi.org/10.19540/j.cnki.cjcmm.20180912.004.

Phan, Q.T., Lin, J., Solis, N.V., Eng, M., Swidergall, M., Wang, F., Li, S., Gaffen, S.L., Chou, T.F., Filler, S.G., 2021. The globular C1q receptor is required for epidermal

growth factor receptor signaling during Candida albicans infection. mBio 12 (6), e0271621. https://doi.org/10.1128/mBio.02716-21.
Ponde, N.O., Lortal, L., Tsavou, A., Hepworth, O.W., Wickramasinghe, D.N., Ho, J., Richardson, J.P., Moyes, D.L., Gaffen, S.L., Naglik, J.R., 2022. Receptor-kinase EGFR-MAPK adaptor proteins mediate the epithelial response to Candida albicans via the cytolytic peptide toxin, candidalysin. J. Biol. Chem. 298 (10), 102419 https://doi.org/10.1016/j.jbc.2022.102419.
Qadir, M.I., Bashir, H., Ahmad, M.H., 2023. Human oropharyngeal candidiasis: from Etiology to current treatment. Crit. Rev. Immunol. 43 (3), 15–24. https://doi.org/10.1615/CritRevImmunol.2023049730.
Qin, Z., Wang, W., Liao, D., Wu, X., Li, X., 2018. UPLC-Q/TOF-MS-Based serum Metabolomics reveals Hypoglycemic effects of Rehmannia glutinosa, Coptis chinensis and their Combination on high-Fat-diet-induced diabetes in KK-ay mice. Int. J. Mol. Sci. 19 (12) https://doi.org/10.3390/ijms19123984.
Rajadurai, S.G., Maharajan, M.K., Veettil, S.K., Gopinath, D., 2021. Comparative efficacy of antifungal agents used in the treatment of oropharyngeal candidiasis among HIV-infected adults: a systematic review and network meta-analysis. J Fungi (Basel) 7 (8). https://doi.org/10.3390/jof7080637.
Rusda, M., Ganis Siregar, M.F., Lelo, A., Ilyas, S., Akbari Ganie, R., Effendi, Y., Zaitun Hasibuan, P.A., Lia Iswara, R.R., 2020. A therapeutic effect of Nigella sativa extract on female Wistar rats vulvovaginal candidiasis model. Med. Glas. 17 (2), 472–476. https://doi.org/10.17392/1112-20.
Seo, Y.S., Yim, M.J., Kim, B.H., Kang, K.R., Lee, S.Y., Oh, J.S., You, J.S., Kim, S.G., Yu, S. J., Lee, G.J., Kim, D.K., Kim, C.S., Kim, J.S., Kim, J.S., 2015. Berberine-induced anticancer activities in FaDu head and neck squamous cell carcinoma cells. Oncol. Rep. 34 (6), 3025–3034. https://doi.org/10.3892/or.2015.4312.
Seoudi, N., Bergmeier, L.A., Drobniewski, F., Paster, B., Fortune, F., 2015. The oral mucosal and salivary microbial community of Behçet's syndrome and recurrent aphthous stomatitis. J. Oral Microbiol. 7, 27150 https://doi.org/10.3402/jom.v7.27150.
Solis, N.V., Filler, S.G., 2012. Mouse model of oropharyngeal candidiasis. Nat. Protoc. 7 (4), 637–642. https://doi.org/10.1038/nprot.2012.011.
Ulian-Benitez, S., Hjelmstad, A.S., Barbosa, G.O., Haghighi, P., 2022. Eosin whole-brain mount staining to analyze neurodegeneration in a fly model of Alzheimer's disease. STAR Protoc. 3 (2), 101377. https://doi.org/10.1016/j.xpro.2022.101377.
Verma, A., Gaffen, S.L., Swidergall, M., 2017. Innate immunity to mucosal Candida infections. J Fungi (Basel) 3 (4). https://doi.org/10.3390/jof3040060.
Vila, T., Sultan, A.S., Montelongo-Jauregui, D., Jabra-Rizk, M.A., 2020. Oral candidiasis: a disease of opportunity. J Fungi (Basel) 6 (1). https://doi.org/10.3390/jof6010015.
Wang, C., Wang, P., Ouyang, H., Wang, J., Sun, L., Li, Y., Liu, D., Jiang, Z., Wang, B., Pan, Z., 2018. Efficacy of traditional Chinese medicine in treatment and prophylaxis of radiation-induced oral mucositis in patients receiving radiotherapy: a randomized controlled trial. Integr. Cancer Ther. 17 (2), 444–450. https://doi.org/10.1177/1534735417725578.
Wang, T., Pan, M., Bao, M., Bu, Q., Yang, R., Yang, Y., Shao, J., Wang, C., Li, N., 2023. Ethyl caffeate combined with fluconazole exhibits efficacy against azole-resistant oropharyngeal candidiasis via the EGFR/JNK/c-JUN signaling pathway. Med. Mycol. 61 (11) https://doi.org/10.1093/mmy/myad114.
Yang, J.S., Wu, C.C., Lee, H.Z., Hsieh, W.T., Tang, F.Y., Bau, D.T., Lai, K.C., Lien, J.C., Chung, J.G., 2016. Suppression of the TNF-alpha level is mediated by Gan-Lu-Yin (traditional Chinese medicine) in human oral cancer cells through the NF-kappa B, AKT, and ERK-dependent pathways. Environ. Toxicol. 31 (10), 1196–1205. https://doi.org/10.1002/tox.22127.
Zorić, N., Kosalec, I., Tomić, S., Bobnjarić, I., Jug, M., Vlainić, T., Vlainić, J., 2017. Membrane of Candida albicans as a target of berberine. BMC Complement Altern. Med. 17 (1), 268. https://doi.org/10.1186/s12906-017-1773-5.